Essential Functional Analysis

Prerequisites for the formalisation of Vern Paulsen's lectures on quantum computing

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Abstract

General facts about complex matrices that the formalised lectures (session `Quantum_Computing_by_VP`) build on: sums, adjoints, unitary matrices and Kronecker products, linear-space isomorphisms, and matrices of fixed dimension as a type, with the Hilbert–Schmidt norm (a normed space for the Hahn–Banach theorem of `HOL-Hahn_Banach`) and the structure of a Euclidean space (so that `HOL-Analysis` applies). Nothing here depends on the lectures.

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The gate names of Isabelle_Marries_Dirac would capture the letters X, Y, Z, H, S, T used as variables in the lectures.

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*

1 Complex numbers and scalar multiples

lemma *cnj-mult-self-cmod*: *cnj z * z = complex-of-real ((cmod z)²)*
⟨*proof*⟩

lemma *smult-smult-mat*:
fixes *A* :: *complex Matrix.mat*
shows *a ·m (b ·m A) = (a * b) ·m A*
⟨*proof*⟩

lemma *zero-smult-mat-carrier*:
fixes *A* :: *complex Matrix.mat*
assumes *A ∈ carrier-mat m k*
shows *0 ·m A = 0m m k*
⟨*proof*⟩

2 The adjoint

The adjoint of an $m \times n$ matrix is $n \times m$.

lemma *dagger-carrier*:
assumes *M ∈ carrier-mat m n*
shows *(M†) ∈ carrier-mat n m*
⟨*proof*⟩

lemma *dagger-mult-diag-inner*:
fixes *Y* :: *complex Matrix.mat*
assumes *Yc*: *Y ∈ carrier-mat p q* **and** *j*: *j < q*
shows *((Y†) * Y) $$ (j,j) = ⟨Matrix.col Y j | Matrix.col Y j⟩*
⟨*proof*⟩

3 Sums of vectors

Finite sums of vectors, entrywise.

definition *vec-sum* :: *nat ⇒ 'c set ⇒ ('c ⇒ complex Matrix.vec) ⇒ complex Matrix.vec* **where**
vec-sum n Js w = Matrix.vec n (λi. sum (λJ. w J $ i) Js)

lemma *vec-sum-dim* [*simp*]: *dim-vec (vec-sum n Js w) = n*
⟨*proof*⟩

lemma *vec-sum-index* [*simp*]: *i < n ⟹ vec-sum n Js w $ i = sum (λJ. w J $ i) Js*
⟨*proof*⟩

lemma *vec-sum-cong*:
assumes *⋀J. J ∈ Js ⟹ w J = w' J*
shows *vec-sum n Js w = vec-sum n Js w'*
⟨*proof*⟩

lemma *mult-mat-vec-vec-sum*:
assumes *Ac*: *A ∈ carrier-mat m n* **and** *fin*: *finite Js*
and *wd*: *⋀J. J ∈ Js ⟹ dim-vec (w J) = n*

shows $A *_v$ *vec-sum n Js w* = *vec-sum m Js* (λJ. $A *_v w J$)
⟨*proof*⟩

lemma *mult-mat-vec-smult-vec*:
fixes A :: *complex Matrix.mat* **and** v :: *complex Matrix.vec*
assumes *Ac*: $A \in$ *carrier-mat m n* **and** *vd*: *dim-vec v* = *n*
shows $A *_v (c \cdot_v v) = c \cdot_v (A *_v v)$
⟨*proof*⟩

4 Products and sums of matrices

lemma *mat-mult-entry*:
fixes A B :: *complex Matrix.mat*
assumes $A \in$ *carrier-mat r s* $B \in$ *carrier-mat s t* $k < r$ $j < t$
shows ($A * B$) $$ (k,j) = ($\sum l$<*s*. A $$ (k,l) * B $$ (l,j))
⟨*proof*⟩

The same with the sum over $\{0.. < q\}$; square matrices that agree on every vector are equal; a non-zero matrix has a non-zero entry.

lemma *mat-mult-index-sum*:
assumes *A-c*: $A \in$ *carrier-mat p q*
and *B-c*: $B \in$ *carrier-mat q s*
and *r-lt*: $r < p$ **and** *c-lt*: $c < s$
shows ($A * B$) $$ (r, c) = ($\sum k$ = *0*..<*q*. A $$ (r, k) * B $$ (k, c))
⟨*proof*⟩

lemma *mat-eq-of-mult-vec-eq*:
fixes M N :: *complex Matrix.mat*
assumes *M-c*: $M \in$ *carrier-mat k k*
and *N-c*: $N \in$ *carrier-mat k k*
and *eq*: $\bigwedge x$. $x \in$ *carrier-vec k* $\Longrightarrow$ $M *_v x = N *_v x$
shows $M = N$
⟨*proof*⟩

lemma *nonzero-entry*:
fixes A :: *complex Matrix.mat*
assumes A: $A \in$ *carrier-mat m k* **and** *nz*: $A \neq 0_m$ *m k*
obtains *r s* **where** $r < m$ $s < k$ A $$ (r,s) $\neq$ *0*
⟨*proof*⟩

An independent frame cancels: if M^*M is invertible, $MAM^* = 0$ forces $A = 0$.

lemma *frame-cancel*:
fixes M A B :: *complex Matrix.mat*
assumes M: $M \in$ *carrier-mat m k* **and** A: $A \in$ *carrier-mat k k* **and** B: $B \in$ *carrier-mat k k*
and *BG*: $B * (M^\dagger * M) = 1_m$ *k* **and** *GB*: $(M^\dagger * M) * B = 1_m$ *k*
and *z*: $M * A * (M^\dagger) = 0_m$ *m m*
shows $A = 0_m$ *k k*
⟨*proof*⟩

lemma *foldr-add-mat-carrier*:
fixes f :: $'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f x \in$ *carrier-mat n n*
shows *foldr* (λx *acc*. $f x$ + *acc*) *xs* (0_m *n n*) $\in$ *carrier-mat n n*
⟨*proof*⟩

lemma *foldr-add-mat-index*:
fixes f :: $'a \Rightarrow$ *complex Matrix.mat*

assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ n\ n$ **and** $i < n$ **and** $j < n$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ n)\ \$\$\ (i,\ j) = sum\text{-}list\ (map\ (\lambda x.\ f\ x\ \$\$\ (i,\ j))\ xs)$
⟨*proof*⟩

lemma *foldr-add-mat-carrier-rect*:
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ p\ q$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ p\ q) \in carrier\text{-}mat\ p\ q$
⟨*proof*⟩

lemma *foldr-add-mat-index-rect*:
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ p\ q$ **and** $i < p$ **and** $j < q$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ p\ q)\ \$\$\ (i,j) = (\sum x \leftarrow xs.\ f\ x\ \$\$\ (i,j))$
⟨*proof*⟩

Carrier-preservation in a different dimension pair.

lemma *foldr-add-carrier-dn*:
fixes $f :: 'a \Rightarrow complex\ Matrix.mat$
assumes $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ d\ n$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ d\ n) \in carrier\text{-}mat\ d\ n$
⟨*proof*⟩

lemma *foldr-add-mat-cong*:
fixes $f\ g :: 'a \Rightarrow complex\ Matrix.mat$
assumes $\bigwedge x.\ x \in set\ xs \Longrightarrow f\ x = g\ x$
shows $foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ z = foldr\ (\lambda x\ acc.\ g\ x + acc)\ xs\ z$
⟨*proof*⟩

Sums over $[0..<N]$, and their entries.

lemma *foldr-upt-add-acc-index*:
fixes $f :: nat \Rightarrow complex\ Matrix.mat$
assumes *carr*: $\forall k < N.\ f\ k \in carrier\text{-}mat\ d\ d$
and *z-c*: $z \in carrier\text{-}mat\ d\ d$
and *r-lt*: $r < d$ **and** *c-lt*: $c < d$
shows $foldr\ (\lambda k\ acc.\ f\ k + acc)\ [0..<N]\ z\ \$\$\ (r,\ c)$
$= (\sum k = 0..<N.\ f\ k\ \$\$\ (r,\ c)) + z\ \$\$\ (r,\ c)$
⟨*proof*⟩

lemma *foldr-upt-add-index*:
fixes $f :: nat \Rightarrow complex\ Matrix.mat$
assumes *carr*: $\forall k < N.\ f\ k \in carrier\text{-}mat\ d\ d$
and *r-lt*: $r < d$ **and** *c-lt*: $c < d$
shows $foldr\ (\lambda k\ acc.\ f\ k + acc)\ [0..<N]\ (0_m\ d\ d)\ \$\$\ (r,\ c)$
$= (\sum k = 0..<N.\ f\ k\ \$\$\ (r,\ c))$
⟨*proof*⟩

lemma *foldr-smult-index*:
assumes *Bs*: $\forall B \in set\ Bs.\ B \in carrier\text{-}mat\ n\ m$ **and** *r*: $r < n$ **and** *s*: $s < m$
shows $foldr\ (\lambda k\ acc.\ c\ k \cdot_m (Bs\ !\ k) + acc)\ [0..<length\ Bs]\ (0_m\ n\ m)\ \$\$\ (r,s)$
$= (\sum k<length\ Bs.\ c\ k * Bs\ !\ k\ \$\$\ (r,s))$
⟨*proof*⟩

4.1 Algebra of *foldr*-sums of matrices

lemma *foldr-add-mat-mult-left*:
fixes $f :: 'a \Rightarrow complex\ Matrix.mat$
assumes *Ac*: $A \in carrier\text{-}mat\ m\ n$ **and** *fc*: $\forall x \in set\ xs.\ f\ x \in carrier\text{-}mat\ n\ p$
shows $A * foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ p) = foldr\ (\lambda x\ acc.\ A * f\ x + acc)\ xs\ (0_m\ m\ p)$
⟨*proof*⟩

lemma *foldr-add-mat-mult-right*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes *Bc*: $B \in$ *carrier-mat n p* **and** *fc*: $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n*
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n) * B =$ *foldr* $(\lambda x\ acc.\ f\ x * B + acc)\ xs\ (0_m\ m\ p)$
⟨*proof*⟩

lemma *foldr-add-mat-dagger*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n*
shows $(foldr\ (\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n))^\dagger = foldr\ (\lambda x\ acc.\ (f\ x)^\dagger + acc)\ xs\ (0_m\ n\ m)$
⟨*proof*⟩

lemma *foldr-add-mat-zero*:
foldr $(\lambda x\ acc.\ (0_m\ m\ n ::$ *complex Matrix.mat*$) + acc)\ xs\ (0_m\ m\ n) = 0_m\ m\ n$
⟨*proof*⟩

lemma *foldr-add-mat-delta*:
fixes $f ::$ *nat* $\Rightarrow$ *complex Matrix.mat*
assumes *fc*: $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat m n* **and** *d*: *distinct xs* **and** *i*: $i \in$ *set xs*
shows *foldr* $(\lambda x\ acc.\ (if\ x = i\ then\ f\ x\ else\ 0_m\ m\ n) + acc)\ xs\ (0_m\ m\ n) = f\ i$
⟨*proof*⟩

lemma *foldr-add-mat-smult-pull*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat* **and** $c :: 'a \Rightarrow$ *complex*
assumes *fc*: $\forall x \in$ *set xs*. $f\ x = c\ x \cdot_m M$ **and** *Mc*: $M \in$ *carrier-mat m n*
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ m\ n) =$ *sum-list* $(map\ c\ xs) \cdot_m M$
⟨*proof*⟩

lemma *foldr-list-as-index*:
fixes $g ::$ *complex Matrix.mat* $\Rightarrow$ *complex Matrix.mat*
shows *foldr* $(\lambda A\ acc.\ g\ A + acc)\ As\ z =$ *foldr* $(\lambda l\ acc.\ g\ (As\ !\ l) + acc)\ [0..<length\ As]\ z$
⟨*proof*⟩

definition *msum* :: *nat* $\Rightarrow$ *nat* $\Rightarrow$ $('a \Rightarrow$ *complex Matrix.mat*$) \Rightarrow$ *'a list* $\Rightarrow$ *complex Matrix.mat* **where**
$msum\ r\ c\ g\ xs = foldr\ (\lambda x\ acc.\ g\ x + acc)\ xs\ (0_m\ r\ c)$

lemma *msum-Nil* [*simp*]: $msum\ r\ c\ g\ [] = 0_m\ r\ c$
⟨*proof*⟩

lemma *msum-Cons* [*simp*]: $msum\ r\ c\ g\ (x\ \#\ xs) = g\ x + msum\ r\ c\ g\ xs$
⟨*proof*⟩

lemma *msum-carrier*: $\forall x \in$ *set xs*. $g\ x \in$ *carrier-mat r c* $\Longrightarrow msum\ r\ c\ g\ xs \in$ *carrier-mat r c*
⟨*proof*⟩

lemma *msum-index*:
assumes $\forall x \in$ *set xs*. $g\ x \in$ *carrier-mat r c* **and** $i < r$ **and** $j < c$
shows $msum\ r\ c\ g\ xs\ \$\$\ (i,j) = (\sum x \leftarrow xs.\ g\ x\ \$\$\ (i,j))$
⟨*proof*⟩

Every *msum* has its stated dimensions; multiplying a sum on the right.

lemma *msum-dims* [*simp*]: *dim-row* $(msum\ r\ c\ g\ xs) = r$ *dim-col* $(msum\ r\ c\ g\ xs) = c$
⟨*proof*⟩

lemma *msum-mult-right*:
assumes $\forall x \in$ *set xs*. $g\ x \in$ *carrier-mat r c* **and** $B \in$ *carrier-mat c p*
shows $msum\ r\ c\ g\ xs * B = msum\ r\ p\ (\lambda x.\ g\ x * B)\ xs$

⟨*proof*⟩

lemma *msum-mult-left*:
assumes $A \in$ *carrier-mat m r* **and** $\forall x \in$ *set xs*. *g x* $\in$ *carrier-mat r c*
shows $A *$ *msum r c g xs* = *msum m c* (λx. $A *$ *g x*) *xs*
⟨*proof*⟩

Sums of matrices over a list: congruence, additivity, scalars, concatenation.

lemma *msum-cong*: ($\bigwedge x$. $x \in$ *set xs* $\Longrightarrow$ *g x* = *h x*) $\Longrightarrow$ *msum r c g xs* = *msum r c h xs*
⟨*proof*⟩

lemma *msum-add*:
assumes *g*: $\forall x \in$ *set xs*. *g x* $\in$ *carrier-mat r c* **and** *h*: $\forall x \in$ *set xs*. *h x* $\in$ *carrier-mat r c*
shows *msum r c* (λx. *g x* + *h x*) *xs* = *msum r c g xs* + *msum r c h xs*
⟨*proof*⟩

lemma *msum-smult*:
assumes *g*: $\forall x \in$ *set xs*. *g x* $\in$ *carrier-mat r c*
shows *msum r c* (λx. *a* $\cdot_m$ *g x*) *xs* = *a* $\cdot_m$ *msum r c g xs*
⟨*proof*⟩

lemma *msum-append*:
assumes *g*: $\forall x \in$ *set xs*. *g x* $\in$ *carrier-mat r c* **and** *g'*: $\forall x \in$ *set ys*. *g x* $\in$ *carrier-mat r c*
shows *msum r c g* (*xs* @ *ys*) = *msum r c g xs* + *msum r c g ys*
⟨*proof*⟩

5 Finite sums and index arithmetic

lemma *sum-list-product-upt*:
($\sum x \leftarrow$ *List.product* [*0*..<*a*] [*0*..<*b*]. *h x*) = ($\sum i$<*a*. $\sum j$<*b*. *h* (*i*,*j*))
⟨*proof*⟩

lemma *sum-delta2*:
fixes *a d b e* :: *nat*
assumes *a*: *a* < *d* **and** *b*: *b* < *e*
shows ($\sum i$<*d*. $\sum j$<*e*. *if i* = *a* $\wedge$ *j* = *b then v else 0*) = *v*
⟨*proof*⟩

A list of 0s and 1s summing to 1 has exactly one 1.

lemma *sum-zero-one-eq-one*:
fixes *D* :: *real list*
assumes *D01*: $\bigwedge i$. *i* < *n* $\Longrightarrow$ *D* ! *i* = *0* $\vee$ *D* ! *i* = *1* **and** *D*: *length D* = *n* **and** *s*: *sum-list D* = *1*
shows $\exists j$<*n*. *D* ! *j* = *1* $\wedge$ ($\forall i$<*n*. *i* $\neq$ *j* $\longrightarrow$ *D* ! *i* = *0*)
⟨*proof*⟩

Concatenating rows of equal length, and the reindexing $\kappa = ik + j$.

lemma *concat-uniform-length*:
assumes $\forall xs \in$ *set L*. *length xs* = *m*
shows *length* (*concat L*) = *length L* $*$ *m*
⟨*proof*⟩

lemma *concat-uniform-nth*:
assumes *uniform*: $\forall xs \in$ *set L*. *length xs* = *m*
and *i-lt*: *i* < *length L*
and *j-lt*: *j* < *m*
shows *concat L* ! (*i* $*$ *m* + *j*) = *L* ! *i* ! *j*
⟨*proof*⟩

lemma *sum-split-prod-atLeastLessThan*:
fixes $f :: nat \Rightarrow 'a::comm\text{-}monoid\text{-}add$
shows $(\sum \kappa = 0..<N * k.\ f\ \kappa) = (\sum i = 0..<N.\ \sum j = 0..<k.\ f\ (i * k + j))$
⟨*proof*⟩

Sums over pairs of indices (i, j) with $i, j < k$.

lemma *msum-pairs-carrier*:
assumes $\bigwedge i\ j.\ i < k \Longrightarrow j < k \Longrightarrow g\ i\ j \in carrier\text{-}mat\ r\ c$
shows $msum\ r\ c\ (\lambda(i,\ j).\ g\ i\ j)\ (List.product\ [0..<k]\ [0..<k]) \in carrier\text{-}mat\ r\ c$
⟨*proof*⟩

lemma *msum-pairs-index*:
assumes *g*: $\bigwedge i\ j.\ i < k \Longrightarrow j < k \Longrightarrow g\ i\ j \in carrier\text{-}mat\ r\ c$ **and** *a*: $a < r$ **and** *b*: $b < c$
shows $msum\ r\ c\ (\lambda(i,\ j).\ g\ i\ j)\ (List.product\ [0..<k]\ [0..<k])\ \$\$\ (a,\ b)$
$= (\sum i{<}k.\ \sum j{<}k.\ g\ i\ j\ \$\$\ (a,\ b))$
⟨*proof*⟩

lemma *msum-pairs-mult-right*:
assumes *g*: $\bigwedge i\ j.\ i < k \Longrightarrow j < k \Longrightarrow g\ i\ j \in carrier\text{-}mat\ r\ c$ **and** *B*: $B \in carrier\text{-}mat\ c\ p$
shows $msum\ r\ c\ (\lambda(i,\ j).\ g\ i\ j)\ (List.product\ [0..<k]\ [0..<k]) * B$
$= msum\ r\ p\ (\lambda(i,\ j).\ g\ i\ j * B)\ (List.product\ [0..<k]\ [0..<k])$
⟨*proof*⟩

lemma *msum-pairs-mult-left*:
assumes *g*: $\bigwedge i\ j.\ i < k \Longrightarrow j < k \Longrightarrow g\ i\ j \in carrier\text{-}mat\ r\ c$ **and** *A*: $A \in carrier\text{-}mat\ m\ r$
shows $A * msum\ r\ c\ (\lambda(i,\ j).\ g\ i\ j)\ (List.product\ [0..<k]\ [0..<k])$
$= msum\ m\ c\ (\lambda(i,\ j).\ A * g\ i\ j)\ (List.product\ [0..<k]\ [0..<k])$
⟨*proof*⟩

lemma *msum-upt-index*:
assumes $\bigwedge x.\ x < m \Longrightarrow g\ x \in carrier\text{-}mat\ r\ c$ **and** $a < r$ **and** $b < c$
shows $msum\ r\ c\ g\ [0..<m]\ \$\$\ (a,\ b) = (\sum x{<}m.\ g\ x\ \$\$\ (a,\ b))$
⟨*proof*⟩

lemma *idx8*: $k < (8::nat) \Longrightarrow k = 0 \vee k = 1 \vee k = 2 \vee k = 3 \vee k = 4 \vee k = 5 \vee k = 6 \vee k = 7$
⟨*proof*⟩

6 Linear maps on a set of matrices

Linearity of a map defined on a set of $n \times n$ matrices (an operator system, from Day 24 on) with values in M_d.

definition *lin-on* :: *nat* ⇒ *nat* ⇒ *complex Matrix.mat set* ⇒ (*complex Matrix.mat* ⇒ *complex Matrix.mat*) ⇒ *bool* **where**
lin-on n d OS Φ ⟷ (∀ *A* ∈ *OS*. Φ *A* ∈ *carrier-mat d d*) ∧ (∀ *A* ∈ *OS*. ∀ *B* ∈ *OS*. Φ (*A* + *B*) = Φ *A* + Φ *B*)
∧ (∀ *A* ∈ *OS*. ∀ *c*. Φ (*c* ·$_m$ *A*) = *c* ·$_m$ Φ *A*)

On a set closed under sums and containing 0, such a map commutes with finite sums.

lemma *msum-closed*:
assumes *S0*: $0_m\ n\ n \in S$ **and** *S*: $\bigwedge A\ B.\ A \in S \Longrightarrow B \in S \Longrightarrow A + B \in S$
and *g*: $\forall x \in set\ xs.\ g\ x \in S$
shows $msum\ n\ n\ g\ xs \in S$
⟨*proof*⟩

lemma *lin-on-msum-closed*:

assumes *L*: *lin-on n d S* Φ **and** *S0*: $0_m\ n\ n \in S$ **and** *S*: $\bigwedge A\ B.\ A \in S \Longrightarrow B \in S \Longrightarrow A + B \in S$
and *g*: $\forall x \in set\ xs.\ g\ x \in S$
shows Φ (*msum n n g xs*) = *msum d d* (λ*x*. Φ (*g x*)) *xs*
⟨*proof*⟩

lemma *lin-on-smult*:
assumes *lin-on n d S F*
shows *lin-on n d S* (λ*X*. *a* $\cdot_m$ *F X*)
⟨*proof*⟩

7 Linear-space isomorphisms

Matrix spaces are represented by carrier predicates on the single type *complex mat*. The record below packages a carrier with its zero, addition and scalar multiplication, so that statements such as $L(\mathcal{H}_A) \otimes L(\mathcal{H}_B) \cong L(\mathcal{H}_A \otimes \mathcal{H}_B)$ (Day 15) can be made as explicit linear isomorphisms with explicit inverses.

record ('*a*, '*v*) *linear-structure* =
ls-carrier :: '*v set*
ls-zero :: '*v*
ls-add :: '*v* ⇒ '*v* ⇒ '*v*
ls-smul :: '*a* ⇒ '*v* ⇒ '*v*

definition *linear-structure-map* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
linear-structure-map V W f ≡
f (*ls-zero V*) = *ls-zero W* ∧
(∀ *x* ∈ *ls-carrier V*. *f x* ∈ *ls-carrier W*) ∧
(∀ *x* ∈ *ls-carrier V*. ∀ *y* ∈ *ls-carrier V*.
f (*ls-add V x y*) = *ls-add W* (*f x*) (*f y*)) ∧
(∀ *c x*. *x* ∈ *ls-carrier V* ⟶
f (*ls-smul V c x*) = *ls-smul W c* (*f x*))

definition *is-isomorphism-between* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
is-isomorphism-between V W f ≡
linear-structure-map V W f ∧
(∃ *g*. *linear-structure-map W V g* ∧
(∀ *x* ∈ *ls-carrier V*. *g* (*f x*) = *x*) ∧
(∀ *y* ∈ *ls-carrier W*. *f* (*g y*) = *y*))

definition *is-isomorphic-to* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ *bool* **where**
is-isomorphic-to V W ≡ ∃ *f*. *is-isomorphism-between V W f*

lemma *is-isomorphism-betweenD-map*:
assumes *is-isomorphism-between V W f*
shows *linear-structure-map V W f*
⟨*proof*⟩

lemma *is-isomorphism-betweenD-inverse*:
assumes *is-isomorphism-between V W f*
obtains *g* **where** *linear-structure-map W V g*
and $\bigwedge x.\ x \in ls\text{-}carrier\ V \Longrightarrow g\ (f\ x) = x$
and $\bigwedge y.\ y \in ls\text{-}carrier\ W \Longrightarrow f\ (g\ y) = y$
⟨*proof*⟩

The example used throughout: $M_{m,n}$ with entrywise addition and scaling.

definition *mat-linear-structure* :: *nat* ⇒ *nat* ⇒ (*complex*, *complex Matrix.mat*) *linear-structure* **where**
mat-linear-structure m n ≡
(| *ls-carrier* = *carrier-mat m n*,
ls-zero = 0_m *m n*,
ls-add = (λ*x y*. *x* + *y*),
ls-smul = (λ*c x*. *c* ·$_m$ *x*) |)

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*
declare *cauchy-schwarz-eq* [*simp del*] *cauchy-schwarz-col* [*simp del*] *eq-from-inner-prod* [*simp del*]

8 Unitary algebra

Library-level consequences of unitarity ($U^\dagger U = I$ and $UU^\dagger = I$ in the library's definition) used throughout; Paulsen's Day 3 characterisations of unitaries are in theory Day03_Unitary_Matrices.

lemma *unitary-square-dim*:
assumes *uA*: *unitary A*
shows *dim-row A* = *dim-col A*
⟨*proof*⟩

lemma *unitary-inv-l-dimcol*:
assumes *uA*: *unitary A*
shows ($A^\dagger$) ∗ *A* = 1_m (*dim-col A*)
⟨*proof*⟩

lemma *unitary-inv-r-dimrow*:
assumes *uA*: *unitary A*
shows *A* ∗ ($A^\dagger$) = 1_m (*dim-row A*)
⟨*proof*⟩

lemma *unitary-carrier-square-col*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-col A*) (*dim-col A*)
⟨*proof*⟩

lemma *unitary-carrier-square-row*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-row A*) (*dim-row A*)
⟨*proof*⟩

lemma *unitary-dagger*:
assumes *uA*: *unitary A*
shows *unitary* ($A^\dagger$)
⟨*proof*⟩

lemma *unitary-mult*:
assumes *UA*: *unitary A*
and *UB*: *unitary B*
and *dimA*: *A* ∈ *carrier-mat n n*
and *dimB*: *B* ∈ *carrier-mat n n*
shows *unitary* (*A* ∗ *B*)
⟨*proof*⟩

9 Kronecker products

Kronecker products of matrices, $A \bigotimes B$: the tensor of identities, the adjoint $(R \otimes T)^* = R^* \otimes T^*$, the mixed-product property $(R_1 \otimes T_1)(R_2 \otimes T_2) = R_1 R_2 \otimes T_1 T_2$ (both from Day 9), and unitarity of tensors of unitaries.

9.1 Tensor of Identities

lemma *tensor-mat-id*:
Tensor.tensor-mat $(1_m\ n)\ (1_m\ m) = 1_m\ (n * m)$
⟨*proof*⟩

9.2 Tensor Adjoint

9.3 Mixed-Product Property

lemma *tensor-mat-mult*:
assumes *dimAC*: *dim-col A* = *dim-row C*
and *dimBD*: *dim-col B* = *dim-row D*
shows *Tensor.tensor-mat A B* ∗ *Tensor.tensor-mat C D* =
Tensor.tensor-mat $(A * C)\ (B * D)$
⟨*proof*⟩

9.4 Corollaries of the Mixed-Product Property

lemma *tensor-adjoint-mult*:
(*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)$) ∗ (*Tensor.tensor-mat A B*) =
Tensor.tensor-mat $((A^\dagger) * A)\ ((B^\dagger) * B)$
⟨*proof*⟩

lemma *tensor-mult-adjoint*:
(*Tensor.tensor-mat A B*) ∗ (*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)$) =
Tensor.tensor-mat $(A * (A^\dagger))\ (B * (B^\dagger))$
⟨*proof*⟩

9.5 Tensor Unitarity

lemma *tensor-mat-unitary-carrier*:
assumes *dimA*: $A \in$ *carrier-mat n n*
and *dimB*: $B \in$ *carrier-mat m m*
and *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary*:
assumes *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-left* [*simp*]:
assumes *uA*: *unitary A*
shows *unitary* (*Tensor.tensor-mat* $(1_m\ n)\ A$)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-right* [*simp*]:
assumes *uA*: *unitary A*

shows *unitary* (*Tensor.tensor-mat A* (1_m *n*))
⟨*proof*⟩

lemma *one-smult-mat*: (*1*::*complex*) $\cdot_m$ *M* = *M*
⟨*proof*⟩

lemma *mult-index-lt*:
fixes *i j n m* :: *nat*
assumes *i* < *n* **and** *j* < *m*
shows *i* ∗ *m* + *j* < *n* ∗ *m*
⟨*proof*⟩

lemma *mult-index-div-mod*:
fixes *i l m* :: *nat*
assumes *l* < *m*
shows (*i* ∗ *m* + *l*) *div m* = *i* (*i* ∗ *m* + *l*) *mod m* = *l*
⟨*proof*⟩

9.6 Tensor products of matrices

lemma *tensor-index-c*:
fixes *A B* :: *complex Matrix.mat*
assumes *A* ∈ *carrier-mat r c B* ∈ *carrier-mat p q i* < *r* ∗ *p j* < *c* ∗ *q* 0 < *c* 0 < *q*
shows *Tensor.tensor-mat A B* $$ (*i*,*j*) = *A* $$ (*i div p*, *j div q*) ∗ *B* $$ (*i mod p*, *j mod q*)
⟨*proof*⟩

lemma *tensor-one-left*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat* (1_m *1*) *A* = *A*
⟨*proof*⟩

lemma *tensor-one-right*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat A* (1_m *1*) = *A*
⟨*proof*⟩

lemma *tensor-mat-add-right*:
fixes *A B C* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat p q* **and** *Cc*: *C* ∈ *carrier-mat p q*
and *c*: 0 < *c* **and** *q*: 0 < *q*
shows *Tensor.tensor-mat A* (*B* + *C*) = *Tensor.tensor-mat A B* + *Tensor.tensor-mat A C*
⟨*proof*⟩

lemma *tensor-mat-add-left*:
fixes *A B C* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat r c* **and** *Cc*: *C* ∈ *carrier-mat p q*
and *c*: 0 < *c* **and** *q*: 0 < *q*
shows *Tensor.tensor-mat* (*A* + *B*) *C* = *Tensor.tensor-mat A C* + *Tensor.tensor-mat B C*
⟨*proof*⟩

lemma *tensor-mat-smult-right*:
fixes *A B* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat p q* **and** *c*: 0 < *c* **and** *q*: 0 < *q*
shows *Tensor.tensor-mat A* (*k* $\cdot_m$ *B*) = *k* $\cdot_m$ *Tensor.tensor-mat A B*
⟨*proof*⟩

lemma *tensor-mat-smult-left*:
fixes A B :: *complex Matrix.mat*
assumes *Ac*: $A \in$ *carrier-mat* r c **and** *Bc*: $B \in$ *carrier-mat* p q **and** c: $0 < c$ **and** q: $0 < q$
shows *Tensor.tensor-mat* $(k \cdot_m A)$ $B = k \cdot_m$ *Tensor.tensor-mat* A B
⟨*proof*⟩

10 Typed-dimension matrix wrapper

The type $('a, 'm :: finite, 'n :: finite)$ *lmat* represents matrices of size $CARD('m) \times CARD('n)$ with entries in $'a$, by typedef of the JNF matrix type at the appropriate dimension.

typedef (**overloaded**) $('a, 'm :: finite, 'n :: finite)$ *lmat* =
$\{M :: 'a$ *Matrix.mat*. *dim-row* $M = CARD('m) \wedge$ *dim-col* $M = CARD('n)\}$
morphisms *unwrap-lm wrap-lm*
⟨*proof*⟩

setup-lifting *type-definition-lmat*

lemma *unwrap-lm-dim-row* [*simp*]:
dim-row (*unwrap-lm* $(M :: ('a, 'm :: finite, 'n :: finite)$ *lmat*$)) = CARD('m)$
⟨*proof*⟩

lemma *unwrap-lm-dim-col* [*simp*]:
dim-col (*unwrap-lm* $(M :: ('a, 'm :: finite, 'n :: finite)$ *lmat*$)) = CARD('n)$
⟨*proof*⟩

lemma *unwrap-lm-carrier* [*simp*]:
unwrap-lm $(M :: ('a, 'm :: finite, 'n :: finite)$ *lmat*$)$
$\in$ *carrier-mat* $(CARD('m))$ $(CARD('n))$
⟨*proof*⟩

11 Typeclass instances

The unique zero matrix of the right shape.

instantiation *lmat* :: (*zero*, *finite*, *finite*) *zero*
begin
lift-definition *zero-lmat* :: $('a, 'm :: finite, 'n :: finite)$ *lmat*
is 0_m $(CARD('m))$ $(CARD('n))$ ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-zero* [*simp*]:
unwrap-lm $(0 :: ('a :: zero, 'm :: finite, 'n :: finite)$ *lmat*$)$
$= 0_m$ $(CARD('m))$ $(CARD('n))$
⟨*proof*⟩

instantiation *lmat* :: (*plus*, *finite*, *finite*) *plus*
begin
lift-definition *plus-lmat* ::
$('a, 'm :: finite, 'n :: finite)$ *lmat* $\Rightarrow$ $('a, 'm :: finite, 'n :: finite)$ *lmat* $\Rightarrow$ $('a, 'm :: finite, 'n :: finite)$ *lmat*
is $(+)$ ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-plus* [*simp*]:
unwrap-lm ((*A* :: (*'a* :: *plus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*) + *B*)
= *unwrap-lm A* + *unwrap-lm B*
⟨*proof*⟩

instantiation *lmat* :: (*uminus*, *finite*, *finite*) *uminus*
begin
lift-definition *uminus-lmat* ::
(*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is *uminus* ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-uminus* [*simp*]:
unwrap-lm (− (*A* :: (*'a* :: *uminus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
= − *unwrap-lm A*
⟨*proof*⟩

instantiation *lmat* :: (*minus*, *finite*, *finite*) *minus*
begin
lift-definition *minus-lmat* ::
(*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is (−) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-minus* [*simp*]:
unwrap-lm ((*A* :: (*'a* :: *minus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*) − *B*)
= *unwrap-lm A* − *unwrap-lm B*
⟨*proof*⟩

Lift the additive structure laws via ⟦⋀*i j*. ⟦*i* < *dim-row ?B*; *j* < *dim-col ?B*⟧ ⟹ *?A* $$ (*i*, *j*) = *?B* $$ (*i*, *j*); *dim-row ?A* = *dim-row ?B*; *dim-col ?A* = *dim-col ?B*⟧ ⟹ *?A* = *?B* on the underlying JNF matrices.

instance *lmat* :: (*semigroup-add*, *finite*, *finite*) *semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*ab-semigroup-add*, *finite*, *finite*) *ab-semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*monoid-add*, *finite*, *finite*) *monoid-add*
⟨*proof*⟩

instance *lmat* :: (*comm-monoid-add*, *finite*, *finite*) *comm-monoid-add*
⟨*proof*⟩

instance *lmat* :: (*group-add*, *finite*, *finite*) *group-add*
⟨*proof*⟩

instance *lmat* :: (*ab-group-add*, *finite*, *finite*) *ab-group-add*
⟨*proof*⟩

12 Real scalar multiplication and Hahn-Banach overloading

HOL's (·) is the abstract real-scalar multiplication used by the Hahn-Banach locales. We overload it on the type (*complex*, *'m*, *'n*) *lmat* as multiplication by the embedded real, lifted from

JNF's $(\cdot_m)$.

overloading
  prod-lmat ≡
    Vector-Space.prod ::
      real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
            ⇒ (*complex*, *'m*, *'n*) *lmat*
begin
definition *prod-lmat* ::
  real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
        ⇒ (*complex*, *'m*, *'n*) *lmat*
where
  prod-lmat r M = *wrap-lm* (*complex-of-real r* $\cdot_m$ *unwrap-lm M*)
end

lemma *unwrap-lm-prod* [*simp*]:
  unwrap-lm (*Vector-Space.prod r* (*M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
    = *complex-of-real r* $\cdot_m$ *unwrap-lm M*
  ⟨*proof*⟩

13 Real-vectorspace locale on the full type

Every (*complex*, *'m*, *'n*) *lmat* with the universe set is a real vector space in HOL/Hahn_Banach's sense.

lemma *vectorspace-UNIV-lmat*:
  Vector-Space.vectorspace (*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

14 The Hilbert-Schmidt norm on (*complex*, *'m*, *'n*) *lmat*

The Hilbert–Schmidt (Frobenius) norm $\|M\| = (\sum_{i,j} |M_{ij}|^2)^{1/2}$, the Euclidean norm of the entries.

definition *hs-norm-lmat* ::
    (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ *real*
where
  hs-norm-lmat M =
    L2-set (λ(*i*, *j*). *cmod* (*unwrap-lm M* $$ (*i*, *j*))) ({*0*..<*CARD*(*'m*)} × {*0*..<*CARD*(*'n*)})

lemma *hs-norm-lmat-nonneg*: *0* ≤ *hs-norm-lmat M*
  ⟨*proof*⟩

Scaling every entry in modulus by $r \geq 0$ scales the norm by r.

lemma *hs-norm-lmat-scale*:
  fixes *M N* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
  assumes ⋀*i j*. *i* < *CARD*(*'m*) ⟹ *j* < *CARD*(*'n*) ⟹
        cmod (*unwrap-lm N* $$ (*i*, *j*)) = *r* ∗ *cmod* (*unwrap-lm M* $$ (*i*, *j*))
    and *0* ≤ *r*
  shows *hs-norm-lmat N* = *r* ∗ *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-smult*:
  hs-norm-lmat (*Vector-Space.prod a M*) = |*a*| ∗ *hs-norm-lmat M*
  ⟨*proof*⟩

lemma *hs-norm-lmat-triangle*:

fixes $M\ N :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat$
shows $hs\text{-}norm\text{-}lmat\ (M + N) \leq hs\text{-}norm\text{-}lmat\ M + hs\text{-}norm\text{-}lmat\ N$
⟨*proof*⟩

15 Subspace, seminorm, normed-vectorspace instances

HS norm is a seminorm on the whole real vector space of $(complex,\ 'm,\ 'n)\ lmat$.

lemma *seminorm-UNIV-hs-norm-lmat*:
$seminorm\ (UNIV :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat\ set)\ hs\text{-}norm\text{-}lmat$
⟨*proof*⟩

HS norm is a (real) norm: it vanishes only on the zero matrix.

lemma *hs-norm-lmat-zero-iff*:
fixes $M :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat$
shows $(hs\text{-}norm\text{-}lmat\ M = 0) \longleftrightarrow (M = 0)$
⟨*proof*⟩

lemma *norm-UNIV-hs-norm-lmat*:
Normed-Space.norm
$(UNIV :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat\ set)\ hs\text{-}norm\text{-}lmat$
⟨*proof*⟩

lemma *normed-vectorspace-UNIV-lmat*:
normed-vectorspace
$(UNIV :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat\ set)\ hs\text{-}norm\text{-}lmat$
⟨*proof*⟩

A complex-linear subspace V of $(complex,\ 'm,\ 'n)\ lmat$ (i.e. nonempty, closed under addition and complex-scalar multiplication) is in particular a real-linear subspace, hence a *subspace* of *UNIV* in HOL/Hahn_Banach's sense.

definition *cmat-subspace-lmat* ::
$(complex,\ 'm :: finite,\ 'n :: finite)\ lmat\ set \Rightarrow bool$
where
$cmat\text{-}subspace\text{-}lmat\ V \equiv$
$V \neq \{\} \wedge$
$(\forall x \in V.\ \forall y \in V.\ x + y \in V) \wedge$
$(\forall x \in V.\ \forall c :: complex.$
$wrap\text{-}lm\ (c \cdot_m unwrap\text{-}lm\ x) \in V)$

lemma *subspace-of-cmat-subspace-lmat*:
fixes $V :: (complex,\ 'm :: finite,\ 'n :: finite)\ lmat\ set$
assumes $cmat\text{-}subspace\text{-}lmat\ V$
shows $Subspace.subspace\ V\ (UNIV :: (complex,\ 'm,\ 'n)\ lmat\ set)$
⟨*proof*⟩

16 Complex scalar multiplication on lmat

Complex scalar multiplication on $(complex,\ 'm,\ 'n)\ lmat$, extending the *Vector-Space.prod* overload (which is only by reals).

definition *cmult-lmat* ::
$complex \Rightarrow (complex,\ 'm :: finite,\ 'n :: finite)\ lmat$
$\Rightarrow (complex,\ 'm,\ 'n)\ lmat$
where
$cmult\text{-}lmat\ c\ M = wrap\text{-}lm\ (c \cdot_m unwrap\text{-}lm\ M)$

lemma *unwrap-lm-cmult* [*simp*]:
unwrap-lm (*cmult-lmat c* (*M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
= *c* ·$_m$ *unwrap-lm M*
⟨*proof*⟩

lemma *cmult-lmat-real*:
cmult-lmat (*complex-of-real a*) *M* = *Vector-Space.prod a M*
⟨*proof*⟩

lemma *cmult-lmat-add*:
fixes *x y* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *cmult-lmat c* (*x* + *y*) = *cmult-lmat c x* + *cmult-lmat c y*
⟨*proof*⟩

lemma *cmult-lmat-scalar-add*:
fixes *x* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *cmult-lmat* (*c* + *d*) *x* = *cmult-lmat c x* + *cmult-lmat d x*
⟨*proof*⟩

lemma *cmult-lmat-scalar-mult*:
fixes *x* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *cmult-lmat* (*c* ∗ *d*) *x* = *cmult-lmat c* (*cmult-lmat d x*)
⟨*proof*⟩

lemma *cmult-lmat-uminus-one*:
fixes *x* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *cmult-lmat* (− *1*) *x* = − *x*
⟨*proof*⟩

lemma *cmult-lmat-i-sq*:
fixes *x* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *cmult-lmat* i (*cmult-lmat* i *x*) = − *x*
⟨*proof*⟩

lemma *hs-norm-lmat-cmult*:
fixes *M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
shows *hs-norm-lmat* (*cmult-lmat c M*) = *cmod c* ∗ *hs-norm-lmat M*
⟨*proof*⟩

The HOL/Hahn_Banach development imports locales also called *norm*, *seminorm* and *continuous*. Hide their short names so that *norm* is the class method from *HOL.Real-Vector-Spaces* and *continuous* the topological predicate.

hide-const (**open**) *Normed-Space.norm*
hide-const (**open**) *Normed-Space.seminorm*
hide-const (**open**) *Function-Norm.continuous*

17 Overview

Nothing in this theory is stated in the notes; it is the bridge that lets the HOL-Analysis theorems on convex sets (separating hyperplanes, compactness) be applied to matrices. (*'a*, *'m*, *'n*) *lmat* is given the typeclass tower *real-vector*, *metric-space*, *real-normed-vector*, *real-inner*, *euclidean-space*, *heine-borel*; for *'a* = *complex* the norm is the Hilbert–Schmidt norm, the inner product its real part, and the dimension is $2\,\mathrm{CARD}('m)\,\mathrm{CARD}('n)$.

18 Real scalar multiplication and *real-vector* instance

For $'a :: real\text{-}vector$, lift element-wise scalar multiplication via $map\text{-}mat\ (scaleR\ a)$.

instantiation *lmat* :: (*real-vector*, *finite*, *finite*) *scaleR*
begin

lift-definition *scaleR-lmat* ::
real ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
is λ*r M*. *map-mat* (λ*x*. *r* ∗$_R$ *x*) *M* ⟨*proof*⟩

instance ⟨*proof*⟩

end

lemma *unwrap-lm-scaleR* [*simp*]:
unwrap-lm ((*r* :: *real*) ∗$_R$ (*M* :: ('*a* :: *real-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*))
= *map-mat* (λ*x*. *r* ∗$_R$ *x*) (*unwrap-lm M*)
⟨*proof*⟩

lemma *unwrap-lm-scaleR-index* [*simp*]:
fixes *M* :: ('*a* :: *real-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *unwrap-lm* (*r* ∗$_R$ *M*) $$ (*i*, *j*) = *r* ∗$_R$ (*unwrap-lm M* $$ (*i*, *j*))
⟨*proof*⟩

instance *lmat* :: (*real-vector*, *finite*, *finite*) *real-vector*
⟨*proof*⟩

19 Entries of sums and differences

lemma *lmat-eq-iff-entries*:
fixes *M N* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
shows *M* = *N* ⟷
(∀ *i j*. *i* < *CARD*('*m*) ⟶ *j* < *CARD*('*n*)
⟶ *unwrap-lm M* $$ (*i*, *j*) = *unwrap-lm N* $$ (*i*, *j*))
⟨*proof*⟩

20 Distance, uniformity, open: *metric-space* instance

Distance is the L2 sum of entry-wise distances, indexed by $\{0..<CARD('m)\} \times \{0..<CARD('n)\}$. Uniformity and *open* are defined via the standard metric-induced filter so that the *open-uniformity* axiom is discharged by reflexivity.

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *dist*
begin

definition *dist-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ *real* **where**
dist-lmat M N = *L2-set* (λ(*i*, *j*). *dist* (*unwrap-lm M* $$ (*i*, *j*)) (*unwrap-lm N* $$ (*i*, *j*)))
({0..<*CARD*('*m*)} × {0..<*CARD*('*n*)})

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *uniformity-dist*

begin

definition *uniformity-lmat* ::
  (('a, 'm :: *finite*, 'n :: *finite*) *lmat* × ('a, 'm, 'n) *lmat*) *filter* **where**
  (*uniformity-lmat* :: (('a, 'm, 'n) *lmat* × ('a, 'm, 'n) *lmat*) *filter*)
    = (*INF* e ∈ {0<..}. *principal* {(x, y). *dist* x y < e})

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *metric-space*
begin

definition *open-lmat* :: ('a, 'm :: *finite*, 'n :: *finite*) *lmat set* ⇒ *bool* **where**
  open-lmat U = (∀ x ∈ U. ∀ F (x', y) *in uniformity*. x' = x ⟶ y ∈ U)

instance ⟨*proof*⟩

end

21 Norm and *real-normed-vector* instance

instantiation *lmat* :: (*real-normed-vector*, *finite*, *finite*) *real-normed-vector*
begin

definition *norm-lmat* :: ('a, 'm :: *finite*, 'n :: *finite*) *lmat* ⇒ *real* **where**
  norm-lmat M = *L2-set* (λ(i, j). *norm* (*unwrap-lm* M $$ (i, j)))
                    ({0..<*CARD*('m)} × {0..<*CARD*('n)})

definition *sgn-lmat* ::
  ('a, 'm :: *finite*, 'n :: *finite*) *lmat* ⇒ ('a, 'm, 'n) *lmat* **where**
  sgn-lmat (M :: ('a, 'm, 'n) *lmat*) = *scaleR* (*inverse* (*norm* M)) M

instance ⟨*proof*⟩

end

22 Real inner product

For 'a :: *real-inner*, the lmat inner product is the entry-wise sum of inner products. When 'a = *complex* with the standard real-inner instance *inner* z w = *Re* (*cnj* z ∗ w), this is the real part of the Hilbert-Schmidt inner product.

instantiation *lmat* :: (*real-inner*, *finite*, *finite*) *real-inner*
begin

definition *inner-lmat* :: ('a, 'm :: *finite*, 'n :: *finite*) *lmat*
                    ⇒ ('a, 'm, 'n) *lmat* ⇒ *real* **where**
  inner-lmat M N = (∑ (i, j) ∈ {0..<*CARD*('m)} × {0..<*CARD*('n)}.
                    inner (*unwrap-lm* M $$ (i, j)) (*unwrap-lm* N $$ (i, j)))

instance ⟨*proof*⟩

end

23 Basis: matrix units carrying a basis vector

definition *lmat-axis* :: *nat* ⇒ *nat* ⇒ *'a* :: *zero*
⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* **where**
lmat-axis i j u = *wrap-lm* (*Matrix.mat* (*CARD*(*'m*)) (*CARD*(*'n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*))

lemma *unwrap-lmat-axis* [*simp*]:
unwrap-lm (*lmat-axis i j u* :: (*'a* :: *zero*, *'m* :: *finite*, *'n* :: *finite*) *lmat*)
= *Matrix.mat* (*CARD*(*'m*)) (*CARD*(*'n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*)
⟨*proof*⟩

Inner product against a single matrix-unit basis vector picks out one matrix entry.

lemma *inner-lmat-axis*:
fixes *u* :: *'a* :: *real-inner*
and *M* :: (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
assumes *ij*: *i* < *CARD*(*'m*) *j* < *CARD*(*'n*)
shows *inner M* (*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*)
= *inner* (*unwrap-lm M* $$ (*i*, *j*)) *u*
⟨*proof*⟩

Two matrix-unit basis vectors are equal iff their entries and positions match (or both vectors are zero).

lemma *lmat-axis-eq-iff*:
fixes *u v* :: *'a* :: *zero*
assumes *i* < *CARD*(*'m* :: *finite*) *j* < *CARD*(*'n* :: *finite*)
and *i'* < *CARD*(*'m*) *j'* < *CARD*(*'n*)
shows (*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*) = *lmat-axis i' j' v*
⟷ (*u* = *v* ∧ *i* = *i'* ∧ *j* = *j'*) ∨ (*u* = *0* ∧ *v* = *0*)
⟨*proof*⟩

Inner product between two matrix-unit basis vectors: 1 if same position and basis component, 0 otherwise.

lemma *inner-lmat-axis-axis*:
fixes *u v* :: *'a* :: *real-inner*
assumes *ij*: *i* < *CARD*(*'m* :: *finite*) *j* < *CARD*(*'n* :: *finite*)
and *ij'*: *i'* < *CARD*(*'m*) *j'* < *CARD*(*'n*)
shows *inner* (*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*) (*lmat-axis i' j' v*)
= (*if i* = *i'* ∧ *j* = *j' then inner u v else 0*)
⟨*proof*⟩

24 Basis set

definition *Basis-lmat-set* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*
where *Basis-lmat-set* = (⋃ *i* ∈ {*0*..<*CARD*(*'m*)}.
⋃ *j* ∈ {*0*..<*CARD*(*'n*)}.
⋃ *u* ∈ *Basis*. {*lmat-axis i j u* :: (*'a*, *'m*, *'n*) *lmat*})

lemma *Basis-lmat-set-finite*:
finite (*Basis-lmat-set* :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

lemma *Basis-lmat-set-nonempty*:
Basis-lmat-set ≠ ({} :: (*'a* :: *euclidean-space*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

lemma *Basis-lmat-orthonormal*:
fixes $u\ v :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
assumes *uB*: $u \in Basis\text{-}lmat\text{-}set$ **and** *vB*: $v \in Basis\text{-}lmat\text{-}set$
shows $inner\ u\ v = (if\ u = v\ then\ 1\ else\ 0)$
⟨*proof*⟩

lemma *axis-mem-Basis-lmat-set*:
fixes $u :: 'a :: euclidean\text{-}space$
assumes *i-lt*: $i < CARD('m :: finite)$
and *j-lt*: $j < CARD('n :: finite)$
and *uB*: $u \in Basis$
shows $(lmat\text{-}axis\ i\ j\ u :: ('a, 'm, 'n)\ lmat) \in Basis\text{-}lmat\text{-}set$
⟨*proof*⟩

lemma *Basis-lmat-perp-imp-zero*:
fixes $x :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
assumes $*$: $\forall u \in Basis\text{-}lmat\text{-}set.\ inner\ x\ u = 0$
shows $x = 0$
⟨*proof*⟩

lemma *zero-inner-Basis-lmat*:
fixes $u :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
shows $inner\ (0 :: ('a, 'm, 'n)\ lmat)\ u = 0$
⟨*proof*⟩

lemma *Basis-lmat-perp-zero*:
fixes $x :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
shows $(\forall u \in Basis\text{-}lmat\text{-}set.\ inner\ x\ u = 0) \longleftrightarrow x = 0$
⟨*proof*⟩

25 Euclidean space instance

instantiation $lmat :: (euclidean\text{-}space, finite, finite)\ euclidean\text{-}space$
begin

definition $Basis\text{-}lmat :: ('a, 'm :: finite, 'n :: finite)\ lmat\ set$ **where**
$Basis\text{-}lmat = Basis\text{-}lmat\text{-}set$

instance ⟨*proof*⟩

end

26 Heine-Borel

instance $lmat :: (euclidean\text{-}space, finite, finite)\ heine\text{-}borel$ ⟨*proof*⟩

27 Dimension

Two distinct (i, j, u) with $u \in Basis\text{-}'a$ give distinct matrix-axis vectors.

lemma *lmat-axis-inj-on-Basis-indices*:
fixes $u\ v :: 'a :: euclidean\text{-}space$
assumes *ij*: $i < CARD('m :: finite)\ j < CARD('n :: finite)$
and *ij'*: $i' < CARD('m)\ j' < CARD('n)$
and *uB*: $u \in Basis$ **and** *vB*: $v \in Basis$
shows $(lmat\text{-}axis\ i\ j\ u :: ('a, 'm, 'n)\ lmat) = lmat\text{-}axis\ i'\ j'\ v$
$\longleftrightarrow i = i' \wedge j = j' \wedge u = v$

⟨*proof*⟩

lemma *DIM-lmat*:
DIM(('*a* :: *euclidean-space*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
= *CARD*('*m*) ∗ *CARD*('*n*) ∗ *DIM*('*a*)
⟨*proof*⟩

28 Component-wise bounds via the L2 norm

lemma *norm-lmat-index-le*:
fixes *M* :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *norm* (*unwrap-lm M* $$ (*i*, *j*)) ≤ *norm M*
⟨*proof*⟩

lemma *dist-lmat-index-le*:
fixes *M N* :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *dist* (*unwrap-lm M* $$ (*i*, *j*)) (*unwrap-lm N* $$ (*i*, *j*)) ≤ *dist M N*
⟨*proof*⟩

29 Convergence: lmat sequence convergence implies entry convergence

lemma *tendsto-unwrap-lm-index*:
fixes *f* :: '*b* ⇒ ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
and *L* :: ('*a*, '*m*, '*n*) *lmat*
assumes *lim*: (*f* ⟶ *L*) *F*
and *ij*: *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows ((λ*x*. *unwrap-lm* (*f x*) $$ (*i*, *j*)) ⟶ *unwrap-lm L* $$ (*i*, *j*)) *F*
⟨*proof*⟩

30 Entries as continuous functions; wrapping

The entry-projection *M* ↦ *unwrap-lm M* $$ (*i*, *j*) is continuous on lmat for valid index (*i*, *j*).

lemma *isCont-unwrap-lm-index*:
fixes *x* :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *ij*: *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *isCont* (λ*M*. *unwrap-lm M* $$ (*i*, *j*)) *x*
⟨*proof*⟩

lemma *continuous-on-unwrap-lm-index*:
fixes *S* :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat set*
assumes *ij*: *i* < *CARD*('*m*) *j* < *CARD*('*n*)
shows *continuous-on S* (λ*M*. *unwrap-lm M* $$ (*i*, *j*))
⟨*proof*⟩

lemma *unwrap-wrap-lm*:
fixes *M* :: *complex Matrix.mat*
assumes *M* ∈ *carrier-mat CARD*('*k*::*finite*) *CARD*('*k*)
shows *unwrap-lm* (*wrap-lm M* :: (*complex*, '*k*, '*k*) *lmat*) = *M*
⟨*proof*⟩

lemma *unwrap-lm-scaleR-mat*:
fixes *R* :: (*complex*, '*k* :: *finite*, '*k*) *lmat*

shows *unwrap-lm* $(u *_R R) =$ *complex-of-real* $u \cdot_m$ *unwrap-lm* R
⟨*proof*⟩